

Scenario	Research question { Testable, IDV+DV }	Hypothesis with reason { If.....then.....because }	Independent variable-What we change { 3 types }	Dependent Variable-What we measure	Control Variable-what we keep same
A student makes paper airplanes of different sizes and flies them to see which one travels the farthest	How does the size of a paper airplane affect the distance it travels ?	If the paper airplane is larger, then it will travel farther because it can generate more lift as larger wings can increase lift and keep the airplane in the air longer.	Small, medium, large airplanes	Distance travelled. measured in metres (m) using a measuring tape from the throwing line to where the airplane lands.	Same paper, same person throwing, same throwing force, same location
Three identical plants are given different amounts of water every day for two weeks . How is growth affected .	How does the amount of water affect plant growth ?	If plants receive more water (up to a suitable amount), then they will grow taller because water is needed for photosynthesis and transport of	50 mL, 100 mL, 150 mL	Plant height measured in centimetres (cm) using a ruler from the base of the stem to the tip of the plant.	Same plant species, same soil, same pot, same sunlight, same duration

		nutrients.			
A toy car is rolled down ramps of different heights . The distance travelled is measured.	How does ramp height affect the distance travelled by a toy car?	If the ramp height is increased, then the car will travel farther because it gains more gravitational potential energy and speed.	Low, medium, high ramp	Distance travelled. measured in metres (m) using a measuring tape from the end of the ramp to where the car stops.	Same toy car, same surface, same release point, no push
An ice cube is placed in sunlight and another ice cube is placed in the shade . The melting time is recorded.	How does the location of an ice cube affect the time taken to melt?	If the ice cube is placed in sunlight, then it will melt faster because it receives more heat energy. Heat energy increases the movement of water molecules, causing ice to melt faster.	Sunlight and shade	Time taken to melt — measured in minutes (min) using a stopwatch from placement until the ice cube completely melts.	Same ice cube size, same container, same starting temperature

Sugar is added to hot water, warm water, and cold water. The time taken to dissolve is measured	How does water temperature affect the time taken for sugar to dissolve?	If the water is hotter, then the sugar will dissolve faster because particles move faster. Higher temperatures increase particle movement and collisions.	Cold, warm and hot water	Time taken to dissolve — measured in seconds (s) using a stopwatch until all sugar disappears	Same amount of sugar, same amount of water, same stirring, same container
A ball is rolled on different surfaces such as carpet, tiles, and grass. From which surface does it roll the fastest.	How does the type of surface affect the distance a ball rolls?	If the ball rolls on tiles, then it will travel farther because tiles produce less friction. Less friction allows the ball to continue moving for a longer distance.	Carpet, tiles and grass.	Distance travelled — measured in metres (m) using a measuring tape from the starting point to where the ball stops.	Same ball, same starting point, same force, level ground
A flashlight is placed at	How does the distance of a	If the flashlight is moved farther	20 cm, 40 cm and 60 cm	Brightness of light spot —	Same flashlight, batteries, room

different distances from a wall. The brightness of the light spot is observed.	flashlight from a wall affect the brightness of the light spot?	from the wall, then the light spot will become dimmer because the light spreads over a larger area.		measured by comparing the brightness on the wall .	lighting, wall colour
Different-sized parachutes are dropped from the same height. The time taken to reach the ground is recorded	How does parachute size affect the time taken to reach the ground?	If the parachute is larger, then it will take longer to reach the ground because it creates greater air resistance. Larger parachutes experience more drag, slowing the fall.	Small, medium and large parachutes	Time taken to reach the ground — measured in seconds (s) using a stopwatch.	Same material, same weight, same drop height, same weather
Students sleep for different numbers of hours before taking a memory test. How are test scores affected.	How does the number of hours of sleep affect memory test scores?	If students sleep longer, then they will score higher because sleep improves memory and concentration.	4 h, 6 h and 8 h	Memory test score — measured as the number of correct answers or marks obtained on the	Same test, same age group, same testing environment,

				memory test.	
A student listens to music at different volume levels while studying and then takes a quiz. How does it affect concentration?	How does music volume affect quiz scores ?	If the student studies with low-volume music, then they may score higher because it is less distracting.	No music, low, medium and high volume	Quiz score — measured as the number of correct answers or marks obtained on the quiz.	Same study material, same study time, same quiz, same room, same student